Confidence Interval on Regression Function

Simultaneous Confidence & Prediction Intervals

1 Confidence Interval on Regression Function

We have learnt ways to find SCI on different linear functions of $\boldsymbol{\beta}$.

One such linear function is

$$\mu^* = \underset{\sim}{x}^{*T}\boldsymbol{\beta},$$

where $\underset{\sim}{x}^*$ is a new observation.

Future observation: $(\, y^*, \, \underset{\sim}{x}^* \,)$:

$$y^* = \underset{\sim}{x}^{*T}\boldsymbol{\beta} + \varepsilon_* = \mu^* + \varepsilon_*.$$

$$E(y^*) = \mu^*.$$

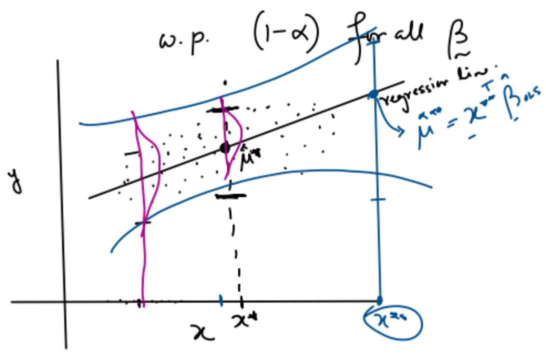
Point estimator of μ^* is:

$$\underset{\sim}{x}^{*T}\hat{\boldsymbol{\beta}}_{\text{OLS}}.$$

Confidence interval

$$\mu^* \;\in\; \underset{\sim}{x}^{*T}\hat{\boldsymbol{\beta}}_{\text{OLS}} \;\pm\; t_{\frac{\alpha}{2}, n-p-1}\sqrt{\frac{\text{RSS}}{n-p-1}\,\underset{\sim}{x}^{*T}(X^TX)^{-1}\underset{\sim}{x}^*}$$

with probability $(1-\alpha)$ for all $\boldsymbol{\beta}$.



$$\begin{cases} \underset{\sim}{x}^{*T}\hat{\boldsymbol{\beta}}_{\text{OLS}} \sim N\big(\underset{\sim}{x}^{*T}\boldsymbol{\beta} = \mu^*, \; \sigma^2\, \underset{\sim}{x}^{*T}(X^TX)^{-1}\underset{\sim}{x}^*\big), \\ \dfrac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-p-1}. \end{cases}$$

$$\Rightarrow \frac{\underset{\sim}{x}^{*T}\hat{\boldsymbol{\beta}}_{\text{OLS}} - \mu^*}{\sqrt{\dfrac{\text{RSS}}{n-p-1}}\sqrt{\underset{\sim}{x}^{*T}(X^TX)^{-1}\underset{\sim}{x}^*}} \sim t_{n-p-1}.$$

We are interested in obtaining a confidence interval for

$$\mu_0 = \underset{\sim}{x}_0^T \boldsymbol{\beta},$$

for all possible values of $\underset{\sim}{x}_0$.

$\underset{\sim}{x}_0$: new observation (unknown).

$$\underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} \sim N\big(\underset{\sim}{x}_0^T \boldsymbol{\beta} = \mu_0,\ \sigma^2\, \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0\big)$$

$$\Rightarrow \frac{\underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} - \mu_0}{\sqrt{\sigma^2\, \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0}} \sim N(0,1)$$

$$\Rightarrow \frac{\underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} - \mu_0}{\sqrt{\dfrac{\text{RSS}}{n-p-1}}\sqrt{\underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0}} \sim t_{n-p-1}.$$

$$\frac{\text{RSS}}{n-p-1} = \hat{\sigma}^2_{\text{MLE}}.$$

$$\sup_{\underset{\sim}{x}_0 \neq \mathbf{0}} \frac{\big|\underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} - \underset{\sim}{x}_0^T \boldsymbol{\beta}\big|}{\hat{\sigma}_{\text{MLE}} \sqrt{\underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0}}.$$

Recall

$$\sup_{\underset{\sim}{x}_0 \neq \mathbf{0}} \frac{\big(\underset{\sim}{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})\big)^2}{\sigma^2\, \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0} = \frac{1}{\sigma^2} (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})^T (X^T X) (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})$$

$$\sim \chi^2_{p+1} \quad \text{[Result ?]}$$

as

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} \sim N\big(\boldsymbol{\beta},\ \sigma^2 (X^T X)^{-1}\big).$$

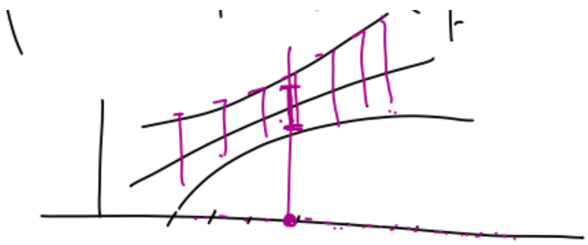
$$\frac{1}{\sigma} (X^T X)^{1/2} (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}) \sim N(\mathbf{0}, I_{p+1}).$$

$$\Rightarrow \sup_{\underset{\sim}{x}_0 \neq \mathbf{0}} \frac{\big\{\underset{\sim}{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})\big\}^2}{\hat{\sigma}^2_{\text{MLE}}\, \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0} \sim F_{p+1,\, n-p-1}.$$

$$P_{\boldsymbol{\beta}}\left(\sup_{\underset{\sim}{x}_0 \neq \mathbf{0}} \frac{\big\{\underset{\sim}{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})\big\}^2}{\hat{\sigma}^2_{\text{MLE}}\, \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0} \ \leq\ F_{\alpha;\, p+1,\, n-p-1} \right) = (1-\alpha)$$

$$\Leftrightarrow\ P_{\boldsymbol{\beta}}\left(\frac{\big|\underset{\sim}{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})\big|}{\hat{\sigma}_{\text{MLE}} \sqrt{\underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0}} \ \leq\ \sqrt{F_{\alpha;\, p+1,\, n-p-1}} \quad \text{for all } \underset{\sim}{x}_0 \neq \mathbf{0} \right) = (1-\alpha)$$

$$\Leftrightarrow\ P_{\boldsymbol{\beta}}\left(\mu_0 = \underset{\sim}{x}_0^T \boldsymbol{\beta} \ \in\ \underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} \ \pm\ \hat{\sigma}_{\text{MLE}} \sqrt{F_{\alpha;\, p+1,\, n-p-1}\, \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0} \quad \forall\, \underset{\sim}{x}_0 \neq \mathbf{0} \right) = (1-\alpha).$$



2 Prediction Interval

For a future point $(\underset{\sim}{x}_0, y_0)$:

$$y_0 = \underset{\sim}{x}_0^T \boldsymbol{\beta} + \varepsilon_0.$$

Assumption: $\varepsilon_0 \sim N(0, \sigma^2)$.

Point estimator of y_0:

$$\underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} = \hat{y}_0.$$

$$\Rightarrow \varepsilon_0 \perp\!\!\!\perp Y, \quad \varepsilon_0 \perp\!\!\!\perp \underset{\sim}{\varepsilon}$$

(consequently $\hat{\boldsymbol{\beta}}_{\text{OLS}}$, RSS).

$$y_0 - \hat{y}_0 = \hat{\varepsilon}_0 = e_0.$$

$$y_0 - \hat{y}_0 = \underset{\sim}{x}_0^T \boldsymbol{\beta} - \underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} + \varepsilon_0.$$

$$\underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} - \underset{\sim}{x}_0^T \underset{\sim}{\boldsymbol{\beta}} - \varepsilon_0 \sim N\big(\underset{\sim}{x}_0^T \boldsymbol{\beta},\ \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0\, \sigma^2 + \sigma^2\big) = \hat{y}_0 - y_0$$
$$\sigma^2 \big(1 + \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0\big).$$

$$\frac{\hat{y}_0 - y_0}{\sigma \sqrt{1 + \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0}} \sim N(0,1) \quad \text{and} \quad \frac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-p-1}$$

(function of $(\hat{\boldsymbol{\beta}}_{\text{OLS}}, \varepsilon_0)$).

And

$$\text{RSS} \perp\!\!\!\perp (\hat{\boldsymbol{\beta}}_{\text{OLS}}, \varepsilon_0).$$

$$\frac{\hat{y}_0 - y_0}{\hat{\sigma}_{\text{MLE}} \sqrt{1 + \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0}} \sim t_{n-p-1}.$$

So the prediction interval for y_0 is

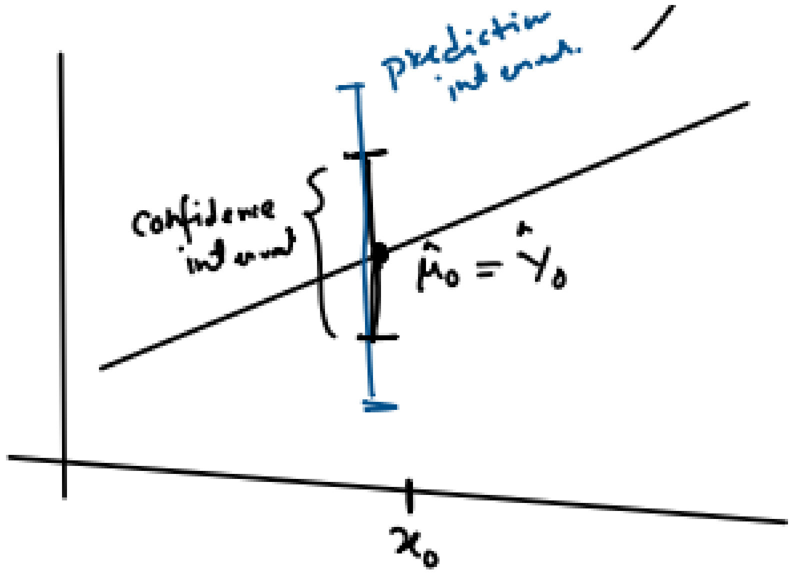
$$f(\text{RSS}, \varepsilon_0, \hat{\boldsymbol{\beta}}_{\text{OLS}}) = f(\varepsilon_0)\, f(\text{RSS}, \hat{\boldsymbol{\beta}}_{\text{OLS}})$$

(function of Y) $\varepsilon_0 \perp\!\!\!\perp \boldsymbol{Y}$.

$$= f(\varepsilon_0)\, f(\text{RSS})\, f(\hat{\boldsymbol{\beta}}_{\text{OLS}}) = f(\text{RSS})\, f(\varepsilon_0,\, \hat{\boldsymbol{\beta}}_{\text{OLS}}).$$

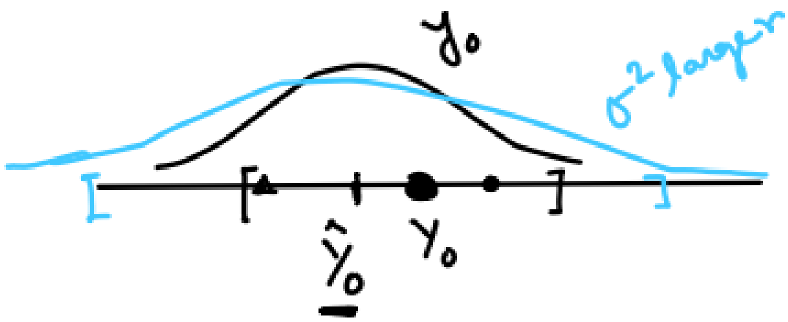
Prediction interval

$$P_{\boldsymbol{\beta}}\left(y_0 \;\in\; \hat{y}_0 \;\pm\; t_{\frac{\alpha}{2}, n-p-1}\, \hat{\sigma}_{\text{MLE}} \sqrt{1 + \underset{\sim}{x}_0^T (X^T X)^{-1} \underset{\sim}{x}_0} \right) = (1 - \alpha).$$



3 Simultaneous Prediction Interval

$$Y_0 = \underset{\sim}{x}_0^T \boldsymbol{\beta} + \varepsilon_0, \qquad \hat{Y}_0 = \underset{\sim}{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}}, \qquad \varepsilon_0 \sim N(0, \sigma^2), \qquad \varepsilon_0 \perp\!\!\!\perp \boldsymbol{\varepsilon}.$$



$$P_{\boldsymbol{\beta}}\left(Y_0 \in \hat{Y}_0 \pm t_{\frac{\alpha}{2};n-p-1}\sqrt{\hat{\sigma}^2_{\text{MLE}}\{1+\underset{\sim}{x}_0^T(X^TX)^{-1}\underset{\sim}{x}_0\}}\right) = (1-\alpha) \quad \forall\,\boldsymbol{\beta},$$

$$P_{\boldsymbol{\beta}}\left(\mu_0 \in \hat{\mu}_0 \pm t_{\frac{\alpha}{2};n-p-1}\sqrt{\hat{\sigma}^2_{\text{MLE}}\,\underset{\sim}{x}_0^T(X^TX)^{-1}\underset{\sim}{x}_0}\right) = (1-\alpha) \quad \forall\,\boldsymbol{\beta},$$

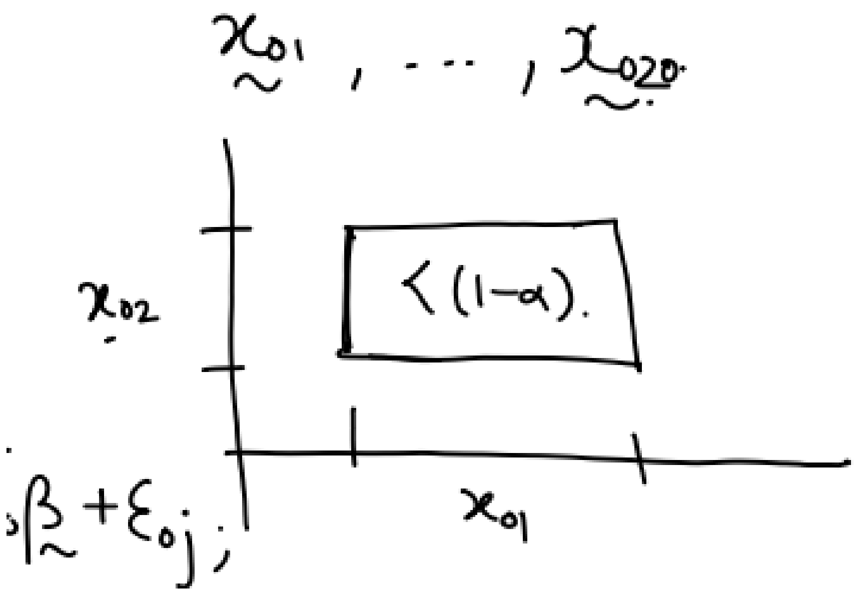
where $\hat{\mu}_0 = \underset{\sim}{x}_0^T\hat{\boldsymbol{\beta}}_{\text{OLS}} = \hat{Y}_0$.

3.1 Generalize prediction interval

- for any future observation $\underset{\sim}{x}_0$;
- k many future Y_0's for same $\underset{\sim}{x}_0$,

$$Y_{0j} = \underset{\sim}{x}_0^T\boldsymbol{\beta} + \varepsilon_{0j}, \qquad j = 1,\ldots,k,$$

and also their combinations $\dfrac{Y_{01}+Y_{02}}{2}$.



Two methods of generalization (by Carlstein).

3.2 Method 1

$$Y_{0j} = \underset{\sim}{x}_0^T \boldsymbol{\beta} + \varepsilon_{0j}, \qquad j = 1, \dots, k.$$

$\underset{\sim}{x}_0$ is unknown.

Assume:

$$\boldsymbol{\varepsilon}_0 = \begin{bmatrix} \varepsilon_{01} \\ \vdots \\ \varepsilon_{0k} \end{bmatrix} \perp\!\!\!\perp \ \boldsymbol{\varepsilon} \quad \text{(and hence is independent of } Y,\ \hat{\boldsymbol{\beta}}_{\text{OLS}},\ \text{RSS).}$$

Define

$$\mathbf{b} = \begin{bmatrix} \boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{\text{OLS}} \\ \boldsymbol{\varepsilon}_0 \end{bmatrix} \sim N_{p+k+1}\left(\mathbf{0},\ \sigma^2 \begin{bmatrix} (X^T X)^{-1} & \mathbf{0} \\ \mathbf{0} & I_k \end{bmatrix}\right).$$

Let

$$W = \begin{bmatrix} (X^T X)^{-1} & \mathbf{0} \\ \mathbf{0} & I_k \end{bmatrix}.$$

Then

$$\frac{1}{\sigma^2}\, \mathbf{b}^T W^{-1} \mathbf{b} \sim \chi^2_{p+k+1}.$$

$$\frac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-p-1} \quad \text{(independent)} \quad \Rightarrow \quad \frac{\mathbf{b}^T W^{-1} \mathbf{b}}{\hat{\sigma}^2_{\text{MLE}}} \sim F_{p+k+1,\, n-p-1}.$$

We know that

$$\mathbf{b}^T W^{-1} \mathbf{b} = \sup_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}^T \mathbf{b})^2}{\mathbf{a}^T W \mathbf{a}} \quad \text{[Result ?].}$$

$$P_{\boldsymbol{\beta}}\left(\frac{\mathbf{b}^T W^{-1} \mathbf{b}}{\hat{\sigma}^2_{\text{MLE}}} \ \le \ F_{\alpha;\, p+k+1,\, n-p-1} \right) = (1-\alpha)$$

$$\Leftrightarrow \ P_{\boldsymbol{\beta}}\left(|\mathbf{a}^T \mathbf{b}| \ \le \ \hat{\sigma}_{\text{MLE}} \sqrt{F_{\alpha;\, p+k+1,\, n-p-1}\, \mathbf{a}^T W \mathbf{a}} \quad \forall\, \mathbf{a} \neq \mathbf{0} \right) = (1-\alpha).$$

What choice of a will give

$$Y_{0j} - \hat{Y}_{0j} = \underset{\sim}{x}_0^T (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{\text{OLS}}) + \varepsilon_{0j}\ ?$$

Ans.

$$\mathbf{a}_j = \begin{bmatrix} \underset{\sim}{x}_0 \\ \mathbf{e}_j \end{bmatrix} \quad \text{where } \mathbf{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow j\text{-th}, \quad j = 1, \dots, k.$$

What choice of a will give

$$\frac{Y_{01}+Y_{02}}{2} - \frac{\hat{Y}_{01}+\hat{Y}_{02}}{2} = M$$

Ans.

$$\mathbf{a}^* = \begin{bmatrix} \underset{\sim}{x}_0 \\ \frac{\mathbf{e}_1+\mathbf{e}_2}{2} \end{bmatrix}.$$

Verification:

$$M = \frac{\underset{\sim}{x}_0^T\boldsymbol{\beta} + \varepsilon_{01} + \underset{\sim}{x}_0^T\boldsymbol{\beta} + \varepsilon_{02}}{2} - \frac{2.\underset{\sim}{x}_0^T\boldsymbol{\beta_{OLS}}}{2},$$

so

$$\frac{Y_{01}+Y_{02}}{2} - \frac{\hat{Y}_{01}+\hat{Y}_{02}}{2} = \underset{\sim}{x}_0^T(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{\mathrm{OLS}}) + \frac{\varepsilon_{01}+\varepsilon_{02}}{2}.$$

$$\text{Thus, } P_{\underset{\sim}{\beta}}\left(\underset{\sim j}{a}^\top \underset{\sim}{b} \in \pm\hat{\sigma}_{\mathrm{MLE}}\sqrt{\underbrace{F_{\alpha;\,p+k+1,\,n-p-1}}_{F_\alpha^*}\underset{\sim j}{a}^\top \bar{W}^{-1}\underset{\sim j}{a}} \quad \begin{matrix} \forall\, j=1,\dots,k \\ \forall\, \underset{\sim 0}{x} \neq \underset{\sim}{0} \end{matrix} \right) \geq (1-\alpha)$$

$$\iff P_{\underset{\sim}{\beta}}\Bigg(Y_{01} \in \hat{Y}_{01} \pm \hat{\sigma}_{\mathrm{MLE}}\sqrt{F_\alpha^*\left\{1 + \underset{\sim 0}{x}^\top (X^\top X)^{-1}\underset{\sim 0}{x}\right\}},\ Y_{02} \in \hat{Y}_{02} \pm \underline{\qquad\qquad\qquad\qquad}, \dots$$

$$, Y_{0k} \in \hat{Y}_{0k} \pm \underbrace{\underline{\qquad\qquad\qquad\qquad\qquad}}_{\forall\, \underset{\sim 0}{x} \neq \underset{\sim}{0}} \Bigg) \geq (1-\alpha).$$

where $\underset{\sim}{a}_j = \begin{bmatrix} \underset{\sim 0}{x} \\ \underset{\sim j}{e} \end{bmatrix}$

For the particular choice $\mathbf{a}_j = \begin{bmatrix} \underset{\sim}{x}_0 \\ \mathbf{e}_j \end{bmatrix}$ one obtains

$$\mathbf{a}_j^T W \mathbf{a}_j = \begin{pmatrix} \underset{\sim}{x}_0^T & \mathbf{e}_j^T \end{pmatrix} \begin{bmatrix} (X^TX)^{-1} & \mathbf{0} \\ \mathbf{0} & I_k \end{bmatrix} \begin{pmatrix} \underset{\sim}{x}_0 \\ \mathbf{e}_j \end{pmatrix} = \underset{\sim}{x}_0^T(X^TX)^{-1}\underset{\sim}{x}_0 + 1.$$

3.3 Method 2

This method treats the uncertainty due to future $\underset{\sim}{x}_0$ and due to randomness of $Y_0 = \underset{\sim}{x}_0^T\boldsymbol{\beta} + \varepsilon_0$ separately.

Recall from simultaneous confidence interval (SCI) of regression mean $\mu_0 = \underset{\sim}{x}_0^T\boldsymbol{\beta}$:

$$P_{\boldsymbol{\beta}}\left(\mu_0 \;\in\; \hat{\mu}_0 \;\pm\; \sqrt{F_{\alpha_1;\,p+1,\,n-p-1}\,\hat{\sigma}^2_{\text{MLE}}\,\underset{\sim}{x}_0^T(X^TX)^{-1}\underset{\sim}{x}_0} \quad \forall\,\underset{\sim}{x}_0 \neq \mathbf{0}\right) = (1-\alpha_1)$$

$$\hat{\mu}_0 = \underset{\sim}{x}_0^T\hat{\boldsymbol{\beta}}_{\text{OLS}}.$$

Now

$$\boldsymbol{\varepsilon}_0 \sim N(\mathbf{0}, \sigma^2 I_k),$$

$$\frac{1}{\sigma^2}\boldsymbol{\varepsilon}_0^T\boldsymbol{\varepsilon}_0 \sim \chi^2_k \quad \text{and} \quad \frac{\text{RSS}}{\sigma^2} \sim \chi^2_{n-p-1}, \quad \text{independent.}$$

$$\frac{\boldsymbol{\varepsilon}_0^T\boldsymbol{\varepsilon}_0}{\hat{\sigma}^2_{\text{MLE}}} \sim F_{k,\,n-p-1}.$$

By Result ?,

$$\boldsymbol{\varepsilon}_0^T\boldsymbol{\varepsilon}_0 = \sup_{\tilde{\mathbf{a}}\neq\mathbf{0}} \frac{(\tilde{\mathbf{a}}^T\boldsymbol{\varepsilon}_0)^2}{\tilde{\mathbf{a}}^T\tilde{\mathbf{a}}}.$$

Using this,

$$P\left(\frac{\boldsymbol{\varepsilon}_0^T\boldsymbol{\varepsilon}_0}{\hat{\sigma}^2_{\text{MLE}}} \;\le\; F_{\alpha_2;\,k,\,n-p-1}\right) = (1-\alpha_2)$$

$$\Leftrightarrow\; P\left(|\tilde{\mathbf{a}}^T\boldsymbol{\varepsilon}_0| \;\le\; \sqrt{F_{\alpha_2;\,k,\,n-p-1}\,\hat{\sigma}^2_{\text{MLE}}\,\tilde{\mathbf{a}}^T\tilde{\mathbf{a}}} \quad \forall\,\tilde{\mathbf{a}}\neq\mathbf{0}\right) = (1-\alpha_2).$$

$$P(A\cap B) = P(A)+P(B)-P(A\cup B) \ge P(A)+P(B)-1 = (1-\alpha_1)+(1-\alpha_2)-1 = 1-(\alpha_1+\alpha_2).$$

Simultaneous coverage

$$P_{\boldsymbol{\beta}}(A\cap B) = P_{\boldsymbol{\beta}}\Bigg(\mu_0+\tilde{\mathbf{a}}^T\boldsymbol{\varepsilon}_0 \;\in\; \hat{\mu}_0 \;\pm\; \left\{\sqrt{F_{\alpha_1;\,p+1,\,n-p-1}\,\underset{\sim}{x}_0^T(X^TX)^{-1}\underset{\sim}{x}_0} + \sqrt{F_{\alpha_2;\,k,\,n-p-1}\,\|\tilde{\mathbf{a}}\|^2}\right\}\hat{\sigma}_{\text{MLE}}$$
$$\forall\,\underset{\sim}{x}_0\neq\mathbf{0},\ \forall\,\tilde{\mathbf{a}}\neq\mathbf{0}\Bigg)$$
$$\ge 1-(\alpha_1+\alpha_2) = 1-\alpha.$$

If we take $\tilde{\mathbf{a}} = \mathbf{e}_j$, $j=1,\ldots,k$, then

$$\mu_0 + \mathbf{e}_j^T\boldsymbol{\varepsilon}_0 = Y_{0j},$$

and the coverage probability satisfies

$$\ge 1-(\alpha_1+\alpha_2) = 1-\alpha \quad (\text{by choosing } \alpha_1+\alpha_2=\alpha).$$

End of notes